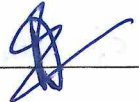


PART-AName of the Student: Kanav BansalSignature of the Student: Kanav BansalRoll No.: 230123029Signature of the Invigilator: **INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	3	4	9	16

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Answer).

Objective $\min(Q(\theta_1, \theta_2)) = \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2$

we need to minimise $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2$ for the arguments θ_1 and θ_2 .

$$\therefore \text{Argmin}_{\theta_1, \theta_2} ((y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2)$$

$$\frac{\partial Q}{\partial \theta_1} = 2(y_1 - \theta_1 - \theta_2)(-1) + 2(y_2 - \theta_1 + 2\theta_2)(-1) + 2(y_3 - 2\theta_1 + \theta_2)(-2)$$

Equating to 0

$$0 = (y_1 - \theta_1 - \theta_2) + (y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2) \quad \text{--- (1)}$$

$$\frac{\partial Q}{\partial \theta_2} = 2(y_1 - \theta_1 - \theta_2)(-1) + 2(y_2 - \theta_1 + 2\theta_2)(2) + 2(y_3 - 2\theta_1 + \theta_2)(1)$$

Equating to 0

$$0 = (-y_1 + \theta_1 + \theta_2) + 2(y_2 - \theta_1 + 2\theta_2) + (y_3 - 2\theta_1 + \theta_2) \quad \text{--- (2)}$$

03

$$\textcircled{1} \Rightarrow 6\theta_1 + \theta_2 = y_1 + y_2 + 2y_3$$

} Normal Equations.

$$\textcircled{2} \Rightarrow -3\theta_1 + 6\theta_2 = y_1 - 2y_2 + y_3$$

$$\therefore \hat{\theta}_2 = \frac{3y_1 - 3y_2 + 4y_3}{13}$$

$$\hat{\theta}_1 = \frac{5y_1 + 8y_2 + 11y_3}{39}$$

$\hat{\theta}_1, \hat{\theta}_2$ are least square estimates of θ_1 and θ_2 .

Let $T: V \rightarrow V$ be a linear transformation. Suppose that $T^2 = T$. Show that T is a projection. (10 marks)

Solution: We need to show that $T^2 = T$ implies T is a projection, i.e., $T^2 = T$. Since $T^2 = T$, we have $T(T(v)) = T(v)$ for all $v \in V$. This means that $T(v)$ is a fixed point of T . Conversely, if v is a fixed point of T , then $T(v) = v$, and thus $T(T(v)) = T(v) = v$. Therefore, T is a projection.

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Answer).

a.) For $Z \sim U(a, b)$ $E[Z] = \frac{a+b}{2}$ $Var(Z) = \frac{(b-a)^2}{12}$

$\therefore E[U_1] = 1/2$ $E[U_2] = 1/2$

$Var[U_1] = \frac{1}{12}$ $Var[U_2] = \frac{1}{12}$

$Cov(U_1, U_2) = 0$ $\because$ They are independent.

$Var(X_1) = Var(U_1) = 1/12$

$Var(X_2) = Var(U_2) = 1/12$

$Var(X_3) = Var(U_1 + U_2) = Var(U_1) + Var(U_2) + 2Cov(U_1, U_2)$
 $= 1/12 + 1/12 + 0 = 1/6$

$Cov(X_1, X_2) = Cov(U_1, U_2) = 0$

$Cov(X_1, X_3) = Cov(U_1, U_1 + U_2) = Cov(U_1, U_1) + Cov(U_1, U_2)$
 $= 1/12$

$Cov(X_2, X_3) = Cov(U_2, U_1 + U_2) = Cov(U_2, U_1) + Cov(U_2, U_2)$
 $= 1/12$

$\therefore$

$$Var(\mathbf{X}) = \begin{bmatrix} 1/12 & 0 & 1/12 \\ 0 & 1/12 & 1/12 \\ 1/12 & 1/12 & 1/6 \end{bmatrix}$$

$$\rho = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$$

03

Using $Corr(X_1, X_2) = \frac{Cov(X_1, X_2)}{\sqrt{Var(X_1)}\sqrt{Var(X_2)}}$

$$b.) \quad X^T \Sigma^{-1} X = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad Z = \text{Normalized}(X)$$

Using $|A - \lambda I| = 0$ to find eigen values and eigen vectors -

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{bmatrix}$$

(A = Σ)

$$|A - \lambda I| = (1-\lambda)[(1-\lambda)^2 - 1/2] = (1-\lambda)[\lambda^2 - 2\lambda + 1/2]$$

$$= (1-\lambda) \lambda (\lambda - 1/\sqrt{2}) \neq 0$$

$\therefore \lambda_1 =$

$$|A - \lambda I| = (1-\lambda) (\lambda - (1+1/\sqrt{2})) (\lambda - (1-1/\sqrt{2}))$$

$$\therefore \lambda_1 = 1+1/\sqrt{2} \quad \lambda_2 = 1 \quad \lambda_3 = 1-1/\sqrt{2}$$

Let the eigen vectors corresponding to these be e_1, e_2, e_3

$$\therefore \text{First Principal Component} = e_1^T Z$$

$$\text{Var}(Y_1) = e_1^T \text{Var}(Z) e_1$$

$$= e_1^T \Sigma e_1$$

$$= \Sigma e_1^T \lambda_1 e_1 = \lambda_1$$

$$\therefore \text{Var}(Y_1) = 1 + \frac{1}{\sqrt{2}}$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Answer

a) i)

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

↓

	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0

↓

	(12)	(34)
(12)	0	
(34)	11	0

↓

	(12 34)
(12 34)	0

Using Complete Linkage

For Complete Linkage

$$d_{(uv)w} = \max(d_{uw}, d_{vw})$$

$$d_{(12)3} = \max\{d_{13}, d_{23}\}$$

$$= \max\{11, 2\} = 11$$

$$d_{(12)4} = \max\{d_{14}, d_{24}\}$$

$$= \max\{5, 3\} = 5$$

$$d_{(12)(34)} = \max\{d_{13}, d_{14}, d_{23}, d_{24}\}$$

$$= \max\{11, 5, 2, 3\} = 11$$

ii.) Using Average Linkage.

For Average Linkage.

$$d_{uw} = \frac{\sum_{i \in U} \sum_{j \in W} d_{ij}}{N_u N_w}$$

$$d_{uw} = \frac{\sum_{i \in U} \sum_{j \in W} d_{ij}}{N_u N_w}$$

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

	(12)	3	4
(12)	0		
3	6.5	0	
4	4	3.5	0

	(12)	(34)
(12)	0	
(34)	5.25	0

	(1234)
(1234)	0

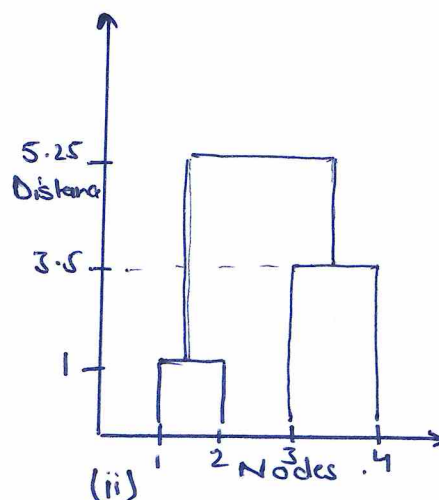
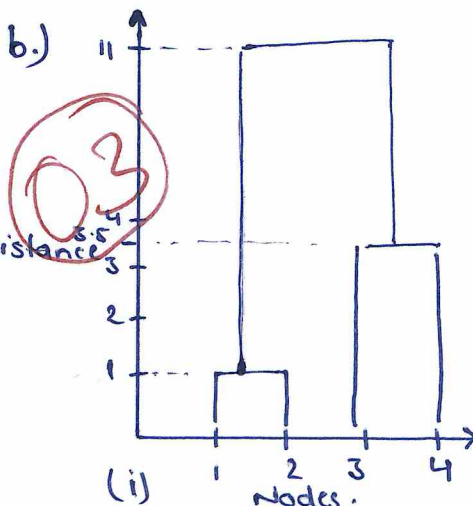
$$d_{(12)3} = \frac{d_{13} + d_{23}}{2 \times 1} = \frac{11 + 2}{2} = 6.5$$

$$d_{(12)4} = \frac{d_{14} + d_{24}}{2 \times 1} = \frac{5 + 3}{2} = 4$$

$$d_{(12)(34)} = \frac{d_{13} + d_{23} + d_{14} + d_{24}}{2 \times 2}$$

$$= \frac{11 + 2 + 5 + 3}{4}$$

$$= \frac{21}{4} = 5.25$$



Dendrograms

For the previous two process we obtained same clusters at each iteration for the two Methods. However in Single, complete, average linkage Single linkage is least stringent, Complete linkage is way stringent and Average linkage lies b/w these two.

ADDITIONAL PAGES FOR ANSWERS

As such the distances b/w clusters reduced faster in the Avg linkage method than the complete linkage method.

